# *The Joy of Computing using Python*

## Assignment 1

1. Which of the following is true about a computer program?

   (a) It is a sequence of instructions

   (b) Instructions that are written in simple english

   (c) There is only one universal programming language

   (d) It is meant for only software developers

   CORRECT ANSWER: (a)

2. Assume you are given two images, each displaying two basic positions of situps. Identify the set of commands to perform an exercise with both images.

   (a) Turn clockwise 90 degree/Turn clockwise -90 degrees

   (b) Move 10 steps/Move -10 steps

   (c) Hide/Show

   (d) point in direction 90/point in direction -90

   CORRECT ANSWER: (c)

3. Choose the best command to be used at the start of your code, to locate the sprite at an initial position every time you play the animation.

   (a) change x to val

   (b) set x to val / set y to val

   (c) point in direction 90

   (d) point towards mouse pointer

   CORRECT ANSWER: (b)

4. Assume the sprite is a ball and predict the output of the following control structure.




   (a) The ball glides to a random position in 1 second

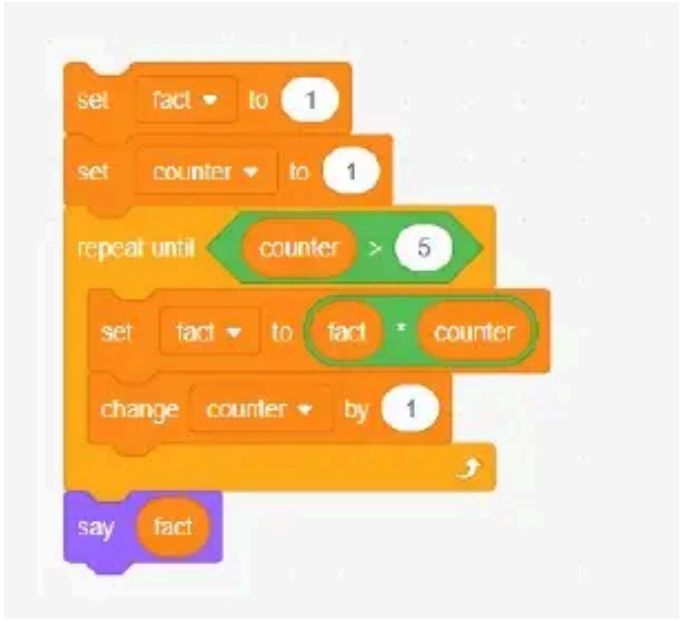
   (b) The ball glides to 10 random positions taking 1 second to reach each position

   (c) The ball glides to a random position and waits there for 10 seconds

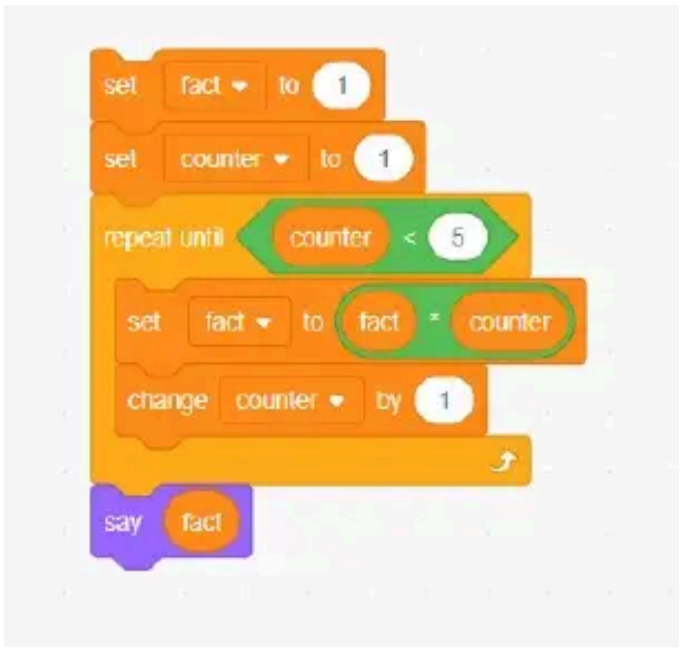
   (d) The ball glides to 10 random positions within 1 second

   CORRECT ANSWER: (b)

5. Pick the snippet that helps the sprite to find the factorial of 5 as output.
   Given:
   Factorial(n)=1 x 2 x 3 x .. x (n-1) x n



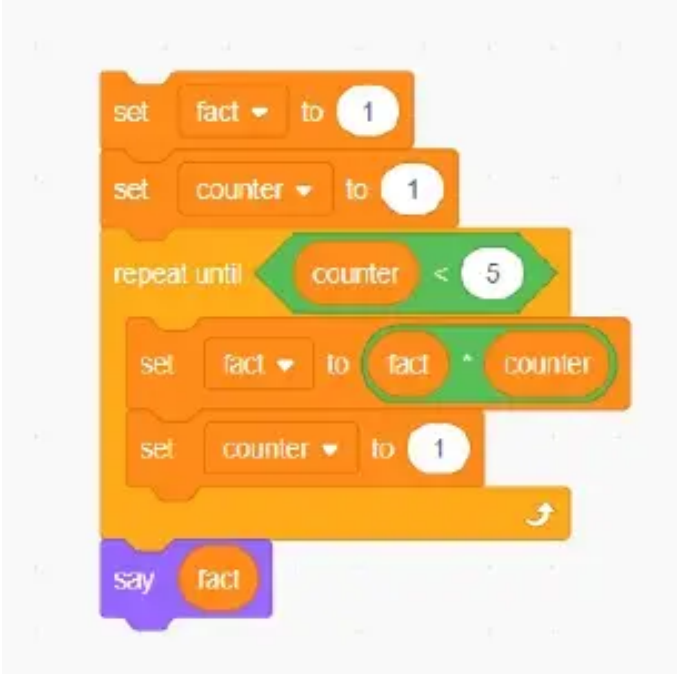

(a)




(b)



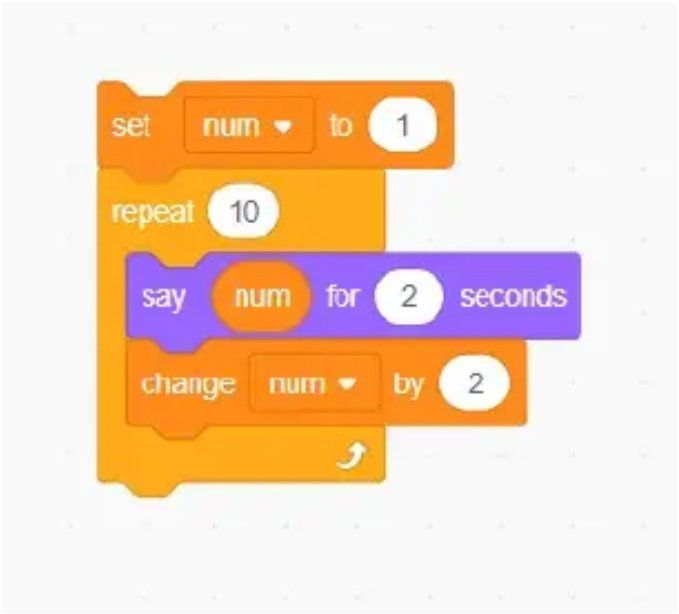

(c)




(d)

CORRECT ANSWER: (a)

6. Predict the sequence of numbers that the sprite recites:



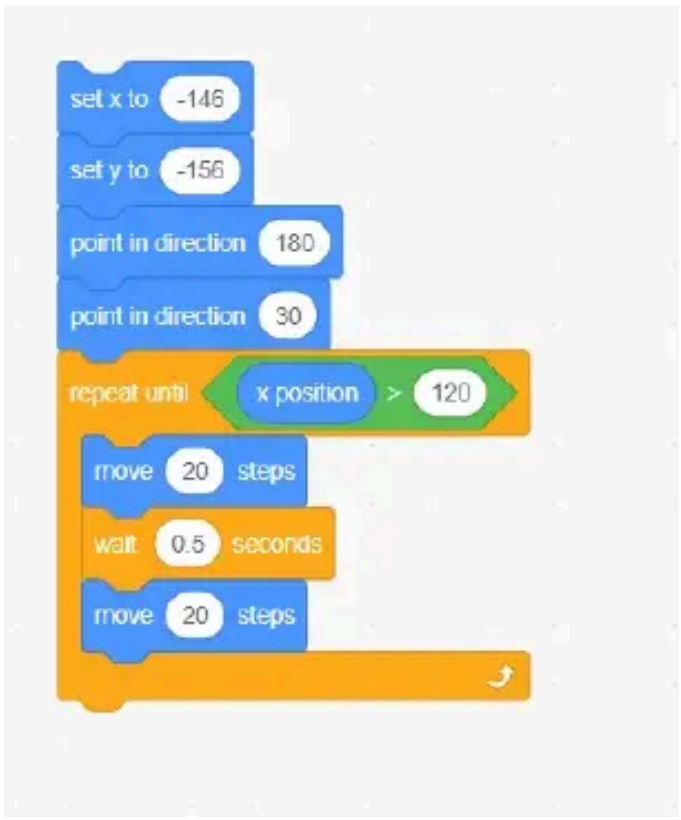

(a) 1, 2, 3, . . . , 19
(b) 1, 3, 5, . . . , 19
(c) 3, 5, 7, . . . , 19
(d) 1, 2, 3, . . . , 20

CORRECT ANSWER: (b)

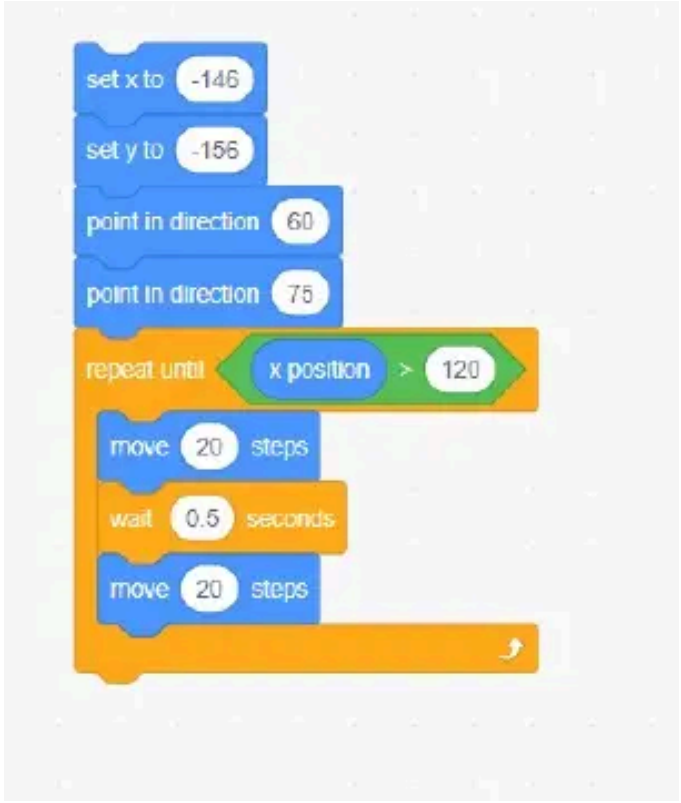
7. Consider a road inclined at an angle of 30° and we have a car sprite to be driven over this road. Pick the code that helps to perform the same. Hint: The initial direction of the sprite is 90°.




(a)




(b)




(c)




(d)

CORRECT ANSWER: (d)

8. Identify the command to communicate across multiple sprites.

   (a) say message
   (b) play sound
   (c) Broadcast message
   (d) touching color

   CORRECT ANSWER: (c)

9. Pick out the scratch library that provides the functionality to switch backdrop?

   (a) Motion

   (b) Control

   (c) Looks

   (d) Sensing

   CORRECT ANSWER: (c)

10. Imagine a Magic wand sprite and predict the output for the following set of instructions.




   (a) The Magic wand flies to a random position takes 1 sec pause and then reaches another random position.

   (b) The Magic wand flies to a random position in 1 sec and after a 0.25 sec pause, it repeats the same until it is stopped.

   (c) The Magic wand reaches all edges of the screen in a uniform pattern

   (d) The Magic wand flies between the top and bottom edges repeatedly.

   CORRECT ANSWER: (b)